

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Project:	Class3 DMR, SH2.0 MAC Address Expansion for SH2.0 Clients	DHF Number:	867214-90003
-----------------	---	--------------------	--------------

Current Revision Change History**As Issued**

Approvers			
Function/Role	Name	Approver Signature	Date¹
ER Manager/Principal Architect	Delroy Smith	See Windchill Doc# D002072203	
Program Manager	Jack Guo	See Windchill Doc# D002072203	
Product Quality Engineer	Asim Ahmad	See Windchill Doc# D002072203	
Technical Reviewer, Sr Software Architect	Cheuw Lee	See Windchill Doc# D002072203	
Solution Test Integration	Sonia Rawal	See Windchill Doc# D002072203	

Approvers: are defined above ft6ewre Added.

¹ Effective Date is the date of last signature.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Table of Contents

1	Purpose.....	3
2	Scope	3
3	Terminology & Abbreviations.....	3
4	Description of System/Solution/Component under test	3
5	Strategy and Approach	7
5.1	New MAC address Impact on SH2.0 System Requirements.....	7
5.2	Test Strategy Details.....	13
5.3	Verification Deliverables	14
6	Impact Assessment.....	14
7	Test Design	15
7.1	SH2.0 Mixed System with new MAC Test Plan – Details	15
7.2	AP Functional Tests with new MAC – SH2.0.....	16
7.3	Mixed System level Test Plan-SH2 – Final Test Details	20
7.4	Sample Size (Mixed Systems)	23
	Sample Plan (Mixed Systems).....	23
7.4.1	AP Test –Sample Size	23
7.4.2	APC Test –Sample Size	23
7.4.3	Large Scale System –Sample Size	24
7.5	SH2.0 System level Test Plan-SH2 – Final Test Details	26
7.6	Sample Size (SH2.0 Systems)	30
7.6.1	AP Test –Sample Size	30
7.6.2	APC Test –Sample Size	31
7.6.3	Large Scale System –Sample Size	31
8	Resources	33
9	Entry Criteria.....	33
10	Exit Criteria	34
11	Suspension and resumption criteria.....	34
12	Defect Management.....	35
13	Test Metrics	35
14	References	35
15	Document Information.....	37
15.1	Document Properties	37
16	Traceability Map – System SRS to Verification Plans	38

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**1 Purpose**

This document plans the management, execution and control of the verification activities of the requirements in scope of this document. Test553.

2 Scope

The scope of this Smart-Hopping2.0 Test Plan is verification of new MAC Address for SH2.0 Clients on Smart Hopping2.0 Mixed System. This verification includes Regression test and new Test associated with the new feature that is defined in the System Requirements Specification for Smart Hopping 2.0, SRS [Ref-01] and Safety Risk Assessment [Ref-02] related to the new feature. The test also included accepted STICRs associated with this project.

3 Terminology & Abbreviations

Terminology & Abbreviations	Description/Definition
AP	Access Point
APC	Access Point Controller
DVIP	Design Verification Iteration Plan
EOL	End of Life
FW	Firmware
GUI	Graphical User Interface
ITS	IntelliVue Telemetry System
IEEE	Institute of Electrical and Electronics Engineers
ITE	Information Technology Equipment
PED	Production Equivalent Devices
PBM	Portable Bedside Monitors
PWM	Patient Worn Monitor, example MX40, Allegro
POE	Power-Over-Ethernet
MAC	Media Access Control
N/A	Not Applicable
RA	Remote Antennas
SU	Sync Units
SRA	Safety Risk Assessment
SRS	System Requirements Specification
STI	System Test Integration
STICR	System Test Integration Change Request
SH	Smart Hopping
SW	Software
VLAN	Virtual Local Area Network
LAN	Local Area Network
UPS	Uninterruptable Power Supply

4 Description of System/Solution/Component under test

Product description:

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

The IntelliVue Telemetry System (ITS), Smart Hopping 2.0 encompasses a number of individual units, which connect together to form a complete method of transporting ambulatory patient data to a central repository for subsequent distribution to clinical staff. An installation comprises:

- Patient Worn Monitors (PWM, MX40, Allegro) and Portable Bedside Monitors (PBM) generically referred to as portable devices, Client devices.
- SH2.0 Client Devices with new MAC address range are supported (Allegro devices, Evoline Bedsides)

The ITS System 1.0 includes the following components:

- Access Points (AP) to which the portable devices have a wireless connection
- Sync Units (SU) which provide a common clock signal for the APs.
- Access Point Controllers (APC) through which data from the APs flows.
- SH2.0 Upgrader Tool (support upgrade of system to D.03)
- POE-Switch, an OEM networking switch component that also provides DC power to APs

The following diagram defines the complete system in terms of medical devices and ITE network infrastructure.

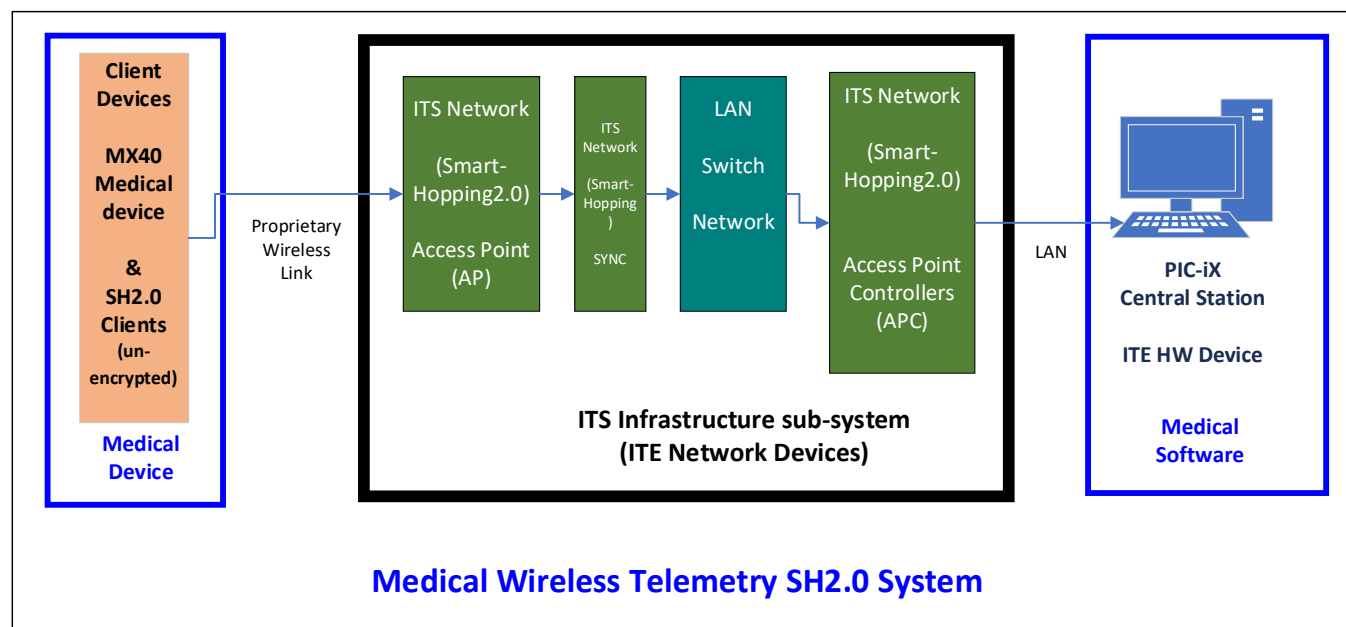


Figure 1: Medical Wireless ITS System

The medical system comprises of Client devices, Patient Worn Monitors or Devices and Medical Central Station. The ITS Network infrastructure comprises of Access Points, Sync network and Access Point Controllers. The components are interconnected by LAN Network Switches. The LAN network can be customer owned where IEEE 80001 applies for risk management.

The focus of the verification is the Smart Hopping 1.0 infrastructure sub-system, the diagram below provides an overall view of this sub-system. This is non-medical and classified as ITE/MME Equipment.

Some major components that comprise the Philips IntelliVue Telemetry System infrastructure are:

1. Access Point Controller (APC) is a physical network device used to manage the operation of the Access Points and provide a gateway between the Access Points (APs) and the Information Center.
 - a. The Serial Console interface provides a serial interface from a service laptop for initial APC configuration.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

- b. The APC Web Interface provides a GUI to the system and configuration management.
2. APCs provide part of the handover function as portable devices move from AP to AP.
3. Access Points (AP) is a network device that provides bi-directional wireless access to the monitoring network for patient-worn transceivers and wireless bedside monitors.
4. Remote Antennas (RA) extends the network coverage area for a core AP.
5. A Synchronization Unit (Sync Unit) transport power and a necessary common clock signal to synchronize all of a system's APs.
6. Power-Over-Ethernet POE-Switch & UPS provide un-interrupted power to the APs.
7. Upgrader Tool is a software application that provides a GUI from a service laptop for system upgrades and other service actions.

Smart Hopping 2.0 Infrastructure Mixed Architecture for SH2.0 Clients

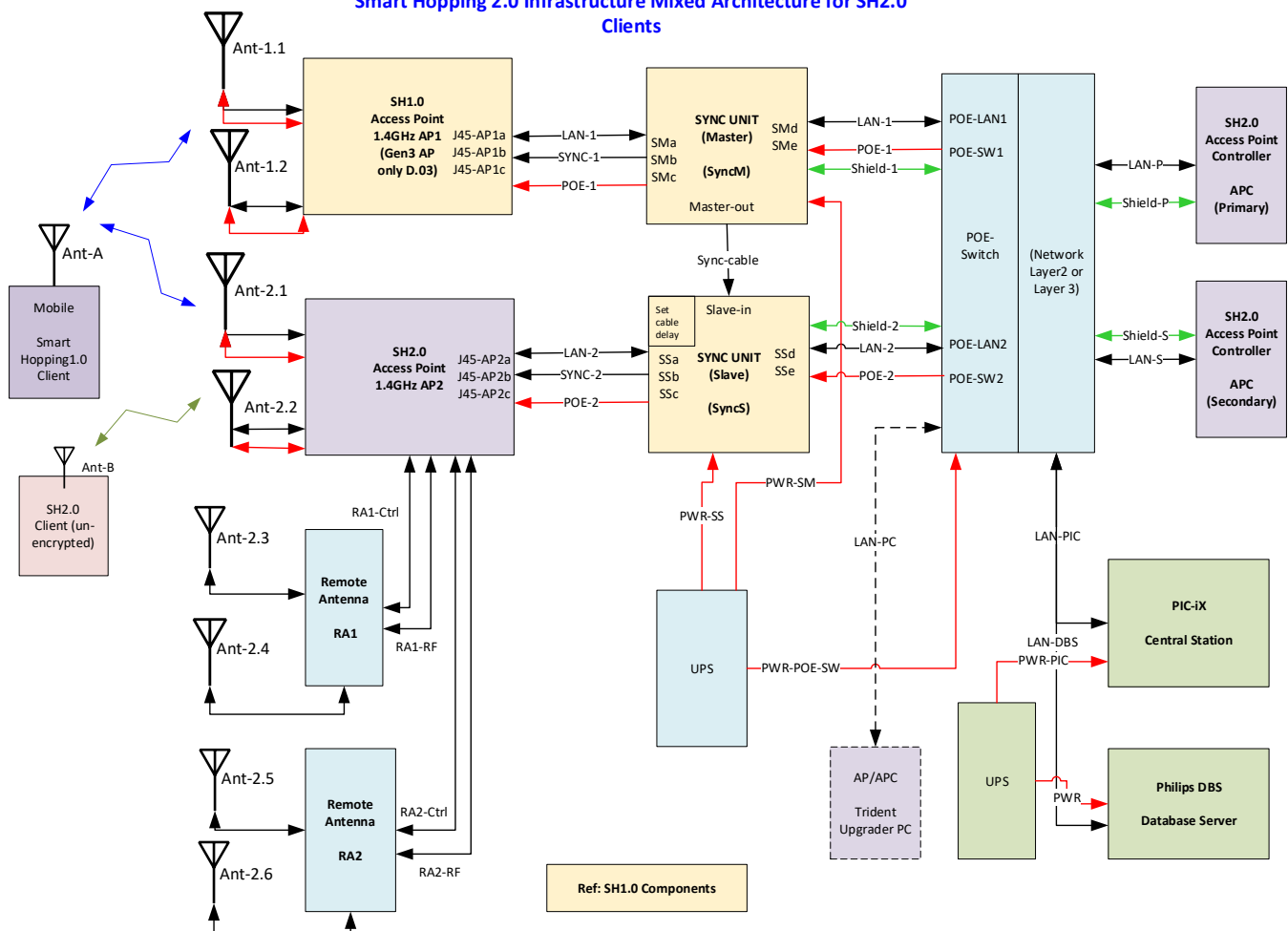


Figure 2: SH 2.0 Subsystems Mixed and Data flow, SH2.0 APC

Note: color used in the above diagram are for "illustration" purposes.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Smart Hopping 2.0 Infrastructure Architecture for SH2.0 Clients

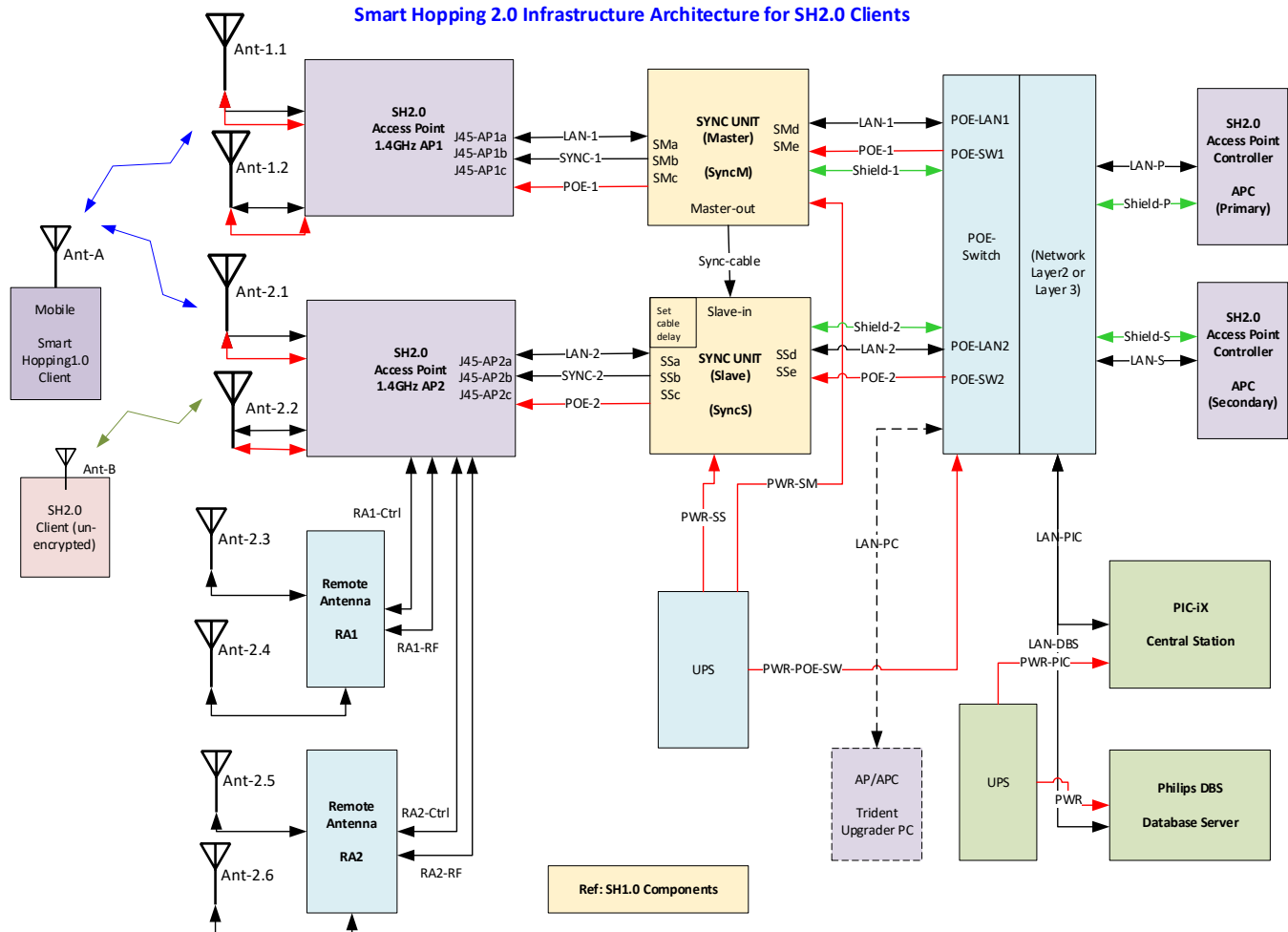


Figure 3: SH 2.0 Subsystems and Data flow, SH2.0 APC (LAPC)

Note: color used in the above diagram are for "illustration" purposes.

The data transport mechanism from the portable devices to the Access Point is a cellular wireless connection using a modified version of the Digital Enhanced Cordless Telephone (DECT) system. Data is delivered to the Central Station for display.

An Ethernet LAN, which includes LAN switches and Routers, is used to interconnect multiple Access Points to one or more APCs. The APC(s) connects on to the PIC(s) and other central station equipment via the same LAN.

The APC(s) also provide system management of the ITS telemetry transport system, via a graphical user interface at a management station.

In the Ethernet feeding each Access Point, there is a UPS, a POE and a Sync Unit (SU), which respectively provide DC power to the Access Point and distribute the common reference clock needed by the wireless system. The APs distribute power from the core AP to the Remote Antennas.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**5 Strategy and Approach****Verification strategy**

- A Master System Verification Report will be created to summarize the results
- STI regression testing will be incorporated into DVIPs.
- The Master System Verification report will include the traceability from requirements to tests generated

Tooling (Consider the following or other description regarding tools)

- STI Lab has all the necessary tools to perform MX40 & SH2.0 Client Test and associated system integration test.
- STI Lab is designed with Smart Hopping1.0 Telemetry Infrastructure, see Fig.2
- STI Lab has suitable ECG, SpO2 Simulators to support the test cases
- STI Lab is designed with LAN Infrastructure to support the regression test, see Fig.2
- Special PWM roaming devices (MX40 & Allegro types) can support automated roaming in R&D and STI test (manual roaming is still required for full effect exposure).

Configurations of system under test

- For STI MX40 shall be tested on the 1.4GHz Smart Hopping1.0 Infrastructure connected to PICiX
- For STI Allegro SH2.0 Client shall be tested on the 1.4GHz Smart Hopping1.0 Infrastructure connected to PICiX
- STI Lab shall include necessary Bedside monitor SH1.0 and SH2.0 Client in the Test and associated system integration test.
- The network shall be configured for both Layer2 and Layer3 (Routed, VLAN network)
- MX40 & Allegro devices under test use the most current versions of hardware/software for production equivalent device (PED).

Test Case to SRS/SRA Mapping (Traceability)**5.1 New MAC address Impact on SH2.0 System Requirements**

			Trident System SRS, AP SW Spec LAPC SW Spec Upgrader Spec		
ID	SH2.0 Feature	Description	New Client MAC Address impact	Test Tag	Test Case
01	Client Roaming	Client Device Roaming between APs	Shall continue to meet User Perceived data loss 90 sec/day within Rssi Threshold (- 67dBm), in SRS (Ref_01) section 4.1.1	<SYS_PERF_01>	Test Case 1 Test Case 1.1
		Client Device Roaming between APs and RAs	includes Bearer handover in SRS (Ref_01) section 4.1.1 and	<SYS_PERF_01>	Test Case 1.2

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

			AP Spec section 4.4 (Ref-03)	SRS_APSW_0403	
			& connection handover included in SRS (Ref_01) section 4.1.1 and AP Spec section 4.4 (Ref-03)	<SYS_PERF_01> SRS_APSW_0404	Test Case 1.3
02	RF Channel Setting	WMTS RF Channel change	RF Channels 1-6 config at APC shall continue to function as defined in SRS (Ref_01) section 3.9.1, APC SW Spec section 4.2.8 (Ref-04), Basic AP Configuration	<SYS_RFCHAN_01> SRS_APP_0270-01	Test Case 2
03	DECT Paging	System Paging MX40 & Allegro Client Device	Shall continue to function as defined in SRS (Ref_01) section 4.1.10, use the Find device function & Standby/Resume shall function per device SRS	<SYS_PER_07><PHASE_MAC> also Defined in MX40 & Allegro SRS	Test Case 3
04	APC Mastership (Premiership) & AP-APC Partnership	APC Mastership (Premiership) Change and AP-APC Partnership change due to system fault	New Alert management shall function as defined in SRS (Ref_01) section 4.5 New Alerts shall not impact the mastership (Premiership) behavior in SRS (Ref_01) section 10.8.2	<SYS_ALERT_02-20><PHASE_MAC> <SYS_ALERT_02-21><PHASE_MAC> <SYS_INSTAL_08-4>	Test Case 4
05	AP-APC Partnership	AP Partnership with APCs change due to system fault	New Alert management shall function as defined in SRS (Ref_01) section 4.5 AP partnership feature shall not be affected, in SRS	<SYS_ALERT_02-20><PHASE_MAC> <SYS_ALERT_02-21><PHASE_MAC> <SYS_INSTAL_08-4>	Test Case 5

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

			(Ref_01) section 10.8.2		
06	Layer 2/3 networks	Solution supports both layer 2 and layer 3 networks	Shall have no impact on network operation, in SRS (Ref_01) section 10.7	<SYS_INSTAL_06-1>	Test Case 6
07	APC Web pages	Individual & Group AP Alert Statistics Web page	No Change to APC Webpage for new PMID Alert, in FSpec (Ref_04) sections 4.2.8, 4.2.11.2	<SRS_APP_0270-07> <SRS_APP_0320-03>	Test Case 7
08	DHCP	APC DHCP Server, update web page	Expand DHCP server range for new MAC address feature Update APC Webpage for DHCP Server, in FSpec (Ref_04) section 4.2.4	<SRS_APP_0230-03>	Test Case 8
		System Recover for AP, covers DHCP server failure	Shall not have no impact on AP recovery as specified in SRS (Ref_01) section 4.3.3	<SYS_SREC_02>	
09	AP Alerts	New AP alerts sent to System for Duplicate PMID Clients	Provide new alert for duplicate PMID Clients to System, in SRS (Ref_01) section 4.5	<SYS_ALERT_02-20><PHASE_MAC>	Test Case 9
10	AP Alerts	Incompatible AP Software Alerts from Primary APC	Provide new alert for Incompatible AP software to System, in SRS (Ref_01) section 4.5	<SYS_ALERT_02-21><PHASE_MAC>	Test Case 10
11	SH2.0 Upgrader	Upgrader supports SH1.0 System	Allow AP Gen3 and RH APC to upgrade, and Block AP Gen1, AP Gen2, in SRS (Ref_01) section 3.9.4, 3.9.5, 3.9.7 and Upgrader Tool SRS (Ref-05)	<SYS_COMP_02> <SYS_COMP_03> <SYS_COMP_04> APC Upgrade: SRS_UPG_0132 <429457_PHASE_AP> SRS_UPG_0133 <429458_PHASE_AP> SRS_UPG_0200 <429466_PHASE_AP> SRS_UPG_0213 <429470_PHASE_AP>	Test Case 11

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

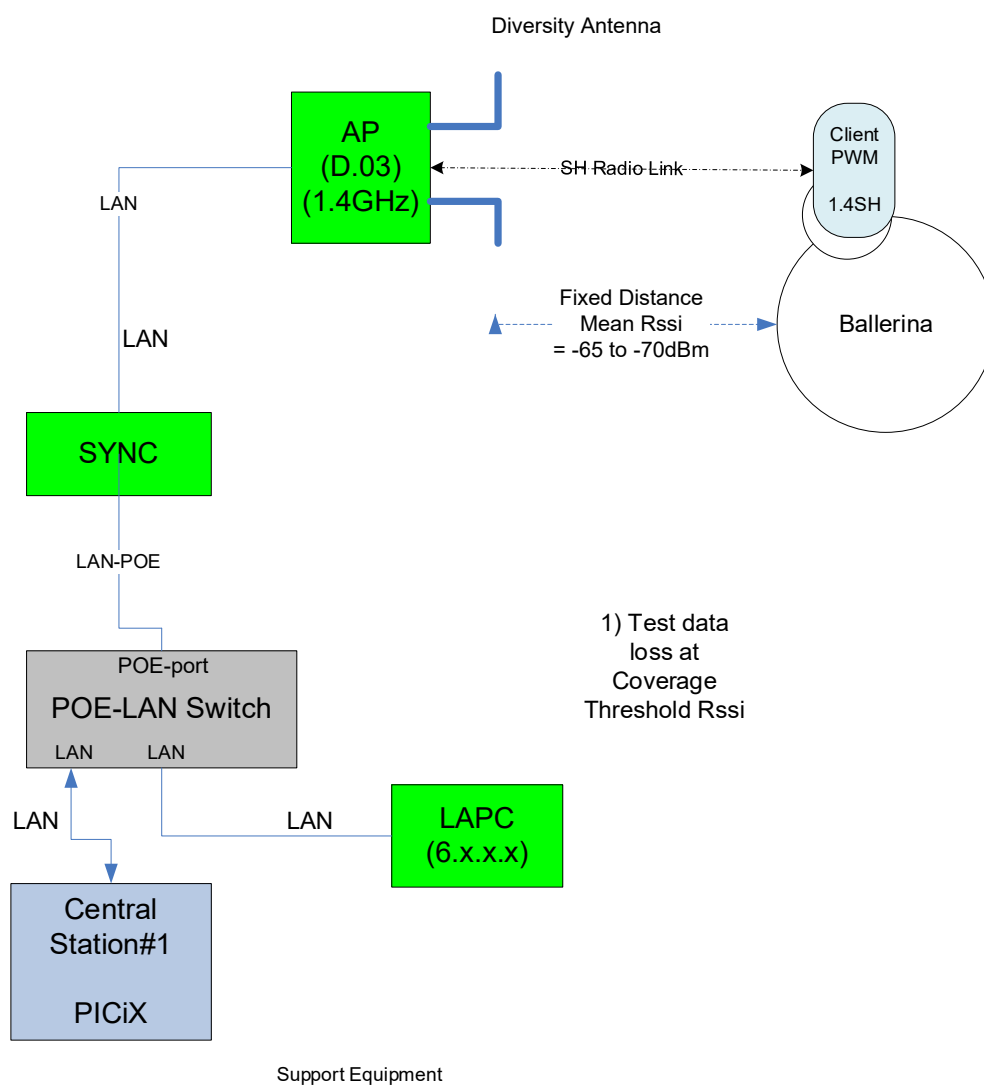
				AP Upgrade: SRS_UPG_0250 <429476_PHASE_AP> SRS_UPG_0260 <429477_PHASE_AP> SRS_UPG_0263 <429480_PHASE_AP>	
12	System Transition from SH1.0 to support SH2.0 Clients with new MAC address	A live system upgrade and parts of the system are not yet upgraded, there are active SH2.0 Clients that can connect to APs that have not been upgraded Incompatible System during AP upgrades with active SH2.0 Clients	PICiX shall not have adverse behavior, such as restart or reboot in SH1.0 system upgrades, in SRS (Ref_01) section 3.9.4, 3.9.5, 3.9.7 and in SRS (Ref_01) section 10.9.5 and Upgrader Tool SRS (Ref-05)	<SYS_COMP_02> <SYS_COMP_03> <SYS_COMP_04> <SYS_INSTAL_11-ID><PHASE_MAC> APC Upgrade: SRS_UPG_0132 <429457_PHASE_AP> SRS_UPG_0133 <429458_PHASE_AP> SRS_UPG_0200 <429466_PHASE_AP> SRS_UPG_0213 <429470_PHASE_AP> AP Upgrade: SRS_UPG_0250 <429476_PHASE_AP> SRS_UPG_0260 <429477_PHASE_AP> SRS_UPG_0263 <429480_PHASE_AP>	Test Case 11
13	Compatibility	Compatibility definition, AP, APC, PICiX, MX40, Focal Point, Remote Antenna, Allego/SH2.0 Clients	Compatibility supported as defined, in SRS (Ref_01) sections 3.9.3, 3.9.4, 3.9.5, 3.9.6, 3.9.7, 3.9.10, 3.9.11	<SYS_COMP_01> <SYS_COMP_02> <SYS_COMP_03> <SYS_COMP_03_1> <SYS_COMP_04> <SYS_COMP_07> <SYS_COMP_08> <SYS_COMP_09>	Test Case 12
14	Export Configuration	Export the system configuration	Shall function as defined in SH2.0 Upgrader Tool SRS, In SRS (Ref_05) section 4.1, 4.6	SRS_UPG_0104 SRS_UPG_0303	Test Case 13
15	Import Configuration	Import the system configuration		SRS_UPG_0105 SRS_UPG_0304	Test Case 14

Note: The Master verification system report will include the final traceability to the SRS and verification, this plan provides the initial traceability between SRS and verification.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Configurations Diagrams Follows:

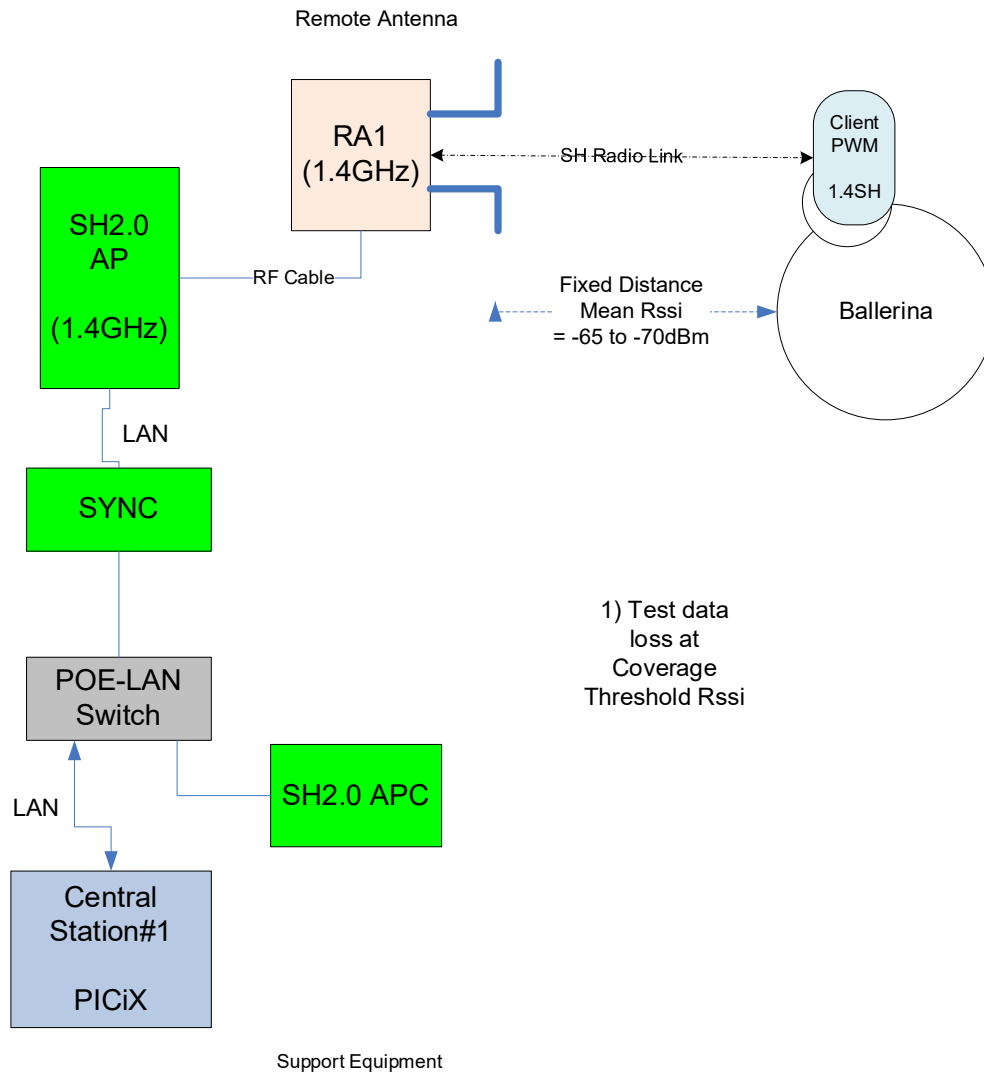
Fig.4 Client PWM on Ballerina, Data Loss Testing with AP (D.03)



Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Fig.7 Client PWM on Ballerina, Data Loss Testing with RA/SH2.0 AP



Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**5.2 Test Strategy Details**

The test strategy is designed to meet the Testing Objectives stated in this document, provide adequate test coverage, and continue to refine the Verification process. The following test models can be considered. Include Test Method where applicable.

The following will be included in Tests:

Test Type	Description	Test Case Creation	Commencement Criteria	Completion Criteria
Review of requirements and specifications	Test engineers will review Requirement documents (SRS, SRA etc.) to provide feedback on the testability and completeness of the product requirements.	N/A	A requirement (or other documentation) is created and declared ready for review.	Updated documents approved.
Integration Test	Integration testing is conducted to evaluate the compliance of a system or component with specified functional requirements.	Tests are designed from specifications.	Integration testing will commence at the successful completion of the build process in the development environment.	Relevant test cases executed. All identified change requests logged per defect tracking process.
Ad-hoc Test	The test team will conduct Ad-hoc functional system/solution testing. Ad-hoc functional system/solution testing will verify new functionality that has been added to the application. Examples of ad-hoc testing include but are not limited to test-a-thons and End2End solution workflows.	N/A	Testing can commence whenever testable items are available.	No specific criteria are needed.
Functional Test	The test team will conduct functional system/solution testing. Functional System/solution testing is focused on product testing with respect to the system/solution. The goal of functional system/solution testing is to verify that the system/solution functions as specified.	Test cases are created from System and Product Requirements, design, use cases/models/workflows, SRA, and based on experiences with existing products/systems/solutions.	Ad-hoc testing for new functionality completed. The application proves functionally stable and can be installed. All functional test cases approved.	Relevant test cases executed. All identified change requests logged per defect tracking process.
Compatibility Test	The test team will conduct functional compatibility testing. Compatibility testing addresses potential issues associated with cross compatibility of 'old' and 'new' components/products in the system/solution.	Test cases are based on compatibility specifications and deployment models.	Compatibility matrix defined, compatibility specifications updated for understanding of feature negotiation across versions of 'old' and 'new' components/products. System/Solution functionally stable and reliability proven.	Relevant test cases executed. All identified change requests logged per defect tracking process.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Test Type	Description	Test Case Creation	Commencement Criteria	Completion Criteria
Performance Test	Performance testing ensures that agreed upon performance thresholds and metrics are maintained in the system.	Test metrics and thresholds are defined by R&D System Architect and included in MSVP.	Testing can commence whenever testable items are available.	Relevant test cases executed. All identified change requests logged per defect tracking process.
Scale Test	Scale test ensures that the core functionality of the system is maintained at the maximum specified limit of the system	Tests are designed from specifications.	Testing can commence whenever testable items are available.	Relevant test cases executed. All identified change requests logged per defect tracking process.
Longevity (aka Continual) Test	The test team will conduct Longevity Testing.	Tests are designed using normal input, output forcing, combination of inputs and robustness.	Longevity testing will commence at the successful completion of the build process in the development environment.	Relevant test cases executed. All identified change requests logged per defect tracking process.
Regression Test	The test team will conduct Regression Testing. Regression testing will verify that existing functionality has not been broken.	Test cases are a full or partial selection to ensure existing functionalities work as specified.	Testing can commence whenever testable items are available.	Relevant test cases executed. All identified change requests logged per defect tracking process.

5.3 Verification Deliverables**SH1.0, SH2.0 Clients**

MX40 Software Transmittal, located in Windchill

MX40 Software C.03.xx, located on eDMR

Allegro Software Transmittal, located in Windchill

Allegro Software 4.0.x.x, located on eDMR

Bedside Monitor with SH2.0 radio (A.05.xx)

Bedside Monitor with SH1.0 radio (FW A.02.54)

Infrastructure SH2.0 Mixed AP

SH2.0 Upgrader Software 6.x.x.x

Gen3 AP Software D.03

SH2.0 AP Software 6.x.x.x

SH2.0 LAPC Software 6.x.x.x

6 Impact Assessment

Design changes impact for the verification activities is as follows:

- Regression Testing i.e. test planning based on regression analysis as defined in this plan

Verification can be an iterative process. Prior to the start of any verification cycle impact assessment is performed, including:

- Retesting any previously failed tests.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

- Verification of defects/ change requests (STICRs)
- Regression Testing.
- Configurations and deployment models

7 Test Design

7.1 SH2.0 Mixed System with new MAC Test Plan – Details

The following verification are performed

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**7.2 AP Functional Tests with new MAC – SH2.0**

Apply to RH AP and SH2.0 AP with support for New MAC address

Test Case 1 AP Summary verification	What to be Tested, Product Level	Final Acceptance Criteria
Test case 1 User Perceived Data Loss Test (Allegro, MX40) on SH1.0 AP (D.03) / LAPC and SH2.0 AP/ LAPC	Measure User Perceived Data Loss with MX40 and Allegro to connected AP, RA 1. Compare Allegro performance to Reference MX40 on Legacy AP (866394) with RA device with new MAC (SW D.03). 2. Compare Allegro performance to Reference MX40 on SH2.0 AP with RA device with new MAC (SW 6.x.x.x). Data Loss Performance: Configure AP to run on the 4 RF Channels (default channels) - a- Use Ballerina to measure Client devices data loss performance. - b- The AP, RA is placed in ceiling or wall in a fixed locations to the test Client device. - c- MX40, Allegro client device is placed on Ballerina, that moves the device through space-time to create multi-path Rf propagation to the AP, RA. - d- Set the Client device Rssi to typically -67 dBm +0/-4 dB by determining a test distance from the AP. - e- Record the Client device Rssi for Rf link to AP, RA. - f- Run the test for 60 minutes and monitor the data loss at the Central Station, data loss rate is estimated from this data ≤ 90 sec/day. - g- Repeat test (b) to (e) with Allegro PWM, Encryption Off - h- The configuration is shown in fig.5 & fig.6 - i- Formal Qualification is done with PP build - j- The Test is done with Allegro and MX40 devices. The following Test is executed by using the STICR process: STICR 1232184. This plan provides the execution details and enhancement as needed to support the evaluation. Note: Alternative test approach can be used to induce Rf fading signal or signal loss, roaming, connection handover and shall be documented in the STICR 1232184 if used. Test includes SH2.0 AP, AP (D.03) with LAPC (6.x.x.x): A1) device Roaming (data loss) within Trident AP and Mixed APs,	As specified in SRS R&D Results reported in STICR 1232184, new MAC RF System Test

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Test Case 1 AP Summary verification	What to be Tested, Product Level	Final Acceptance Criteria
	A2) device density, A3) connection handover (Trident SH2.0 APs and Mixed SH1.0 APs, SH2.0 APs) A4) DECT Paging Test A5) Duplicate PMID test	

Table-5 Functional Test Cases Continue-1

SH1.0 AP with RA Functionality Tests (Test Case 1.3)				
Test case 1.3	Preconditions	Actions	Expected behavior	Owner
Bearer Handover test #4 (AP Only)	Special Allegro PWM (with new MAC, Encryption Off) to force automatic handover Normal system with 1 AP with 0 RA (Core AP only)	1) Connect 1 Special PWM (On Core AP) 2) Disconnect RA1 & RA2 from Core AP 3) Special PWM will randomly roam across Core AP time-slots 4) run test for at least 30 minutes Note: This is a software type test (1 sample)	The average data loss for bearer handover: 99% bearer handover without dropout (at central station) over a 10 minute period.	RTX RTX3481 Software Test Specification, Test Case TC17
Bearer Handover test #5 (AP & RA's)	Special Allegro PWM (with new MAC, Encryption Off) to force automatic handover Normal system with 1 AP with 2 RA	1) Connect 1 Special PWM (On Core AP) 2) Connect antennas on RA1 & RA2 3) Special PWM will randomly roam across all elements, RA1, RA2 and Core 4) run test for at least 30 minutes Note: This is a software type test (1 sample)	The average data loss for bearer handover: 99% bearer handover without dropout (at central station) over a 10 minute period.	RTX RTX3481 Software Test Specification, Test Case TC17
Connection Handover test #5	Special Allegro PWM (with new MAC, Encryption Off) to force automatic handover	1) Connect 1 Special PWM (On Core AP) 2) Connect antennas on RA1 & RA2 3) Special PWM (with new MAC) will randomly roam across all elements, RA1, RA2 and Core 4) run test for at least 30 minutes, connected to PICiX	The average data loss for bearer handover: 90% connection handover without dropout (at central station) over a 10 minute period.	RTX RTX3481 Software Test Specification, Test Case TC7, TC8

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

	Normal system with 1 AP with 2 RA	Note: This is a software type test (1 sample)		
AP load test #3	Normal system with 1 AP and 2 RF Channel. The PICiX should support 32beds	This is software test, 1 AP Samples is required 1) Connect 32 Allegro PWM, Encryption Off, to AP 2) Enable 2 RF channels 3) Run test for 24 hours	1) There should be waveform data for all 32 devices. (monitor for approx 24 hours) 2) data loss over 24 hours, < 90 sec/day per client device 3) No loss of association between Client devices and Central Station	R&D, RTX RTX3481 Software, RTX AP Software Test: TC22, TC54
AP Load test #4	Normal system with 1 AP and 2 RF Channels. The PICiX should support 32beds	This is software test, 1 AP Samples is required 1) Turn on 32 Allegro PWMs at the same time. 2) Turn off all 32 Allegro PWMs, Encryption Off, at the same time. 3) After 1 min, turn all 32 Allegro PWM on again. 4) Repeat (1) to (3) 15 times. This creates 480 association events. 5) This is an attribute test, with 95% confidence/99% prop and accepting following statistics 299 events with 0 failure 473 events with 1 failure	1-2) All Allegro PWM should establish connections and start streaming waveform. 3-4) All Allegro PWM should establish connections and start streaming waveform. 5) Attribute statistics are met	R&D, RTX RTX3481 Software, RTX AP Software Test: TC22
For tests with Allegro device auto roamer Enabling Special Allegro device in automatic roaming mode 1. For each Allegro device, enable auto roam with pwminterface commands: a. Device.devmode=1 b. Test.cts.handover=conn c. Power cycle and let the devices run 1. Review RFDA.log from PICiX 2. Record the following information from each device a. Start Time, End Time, Duration b. Packet Loss Count c. Review Device Location via PICiX log viewer application for roaming events				

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Table-6 Functional Test Cases Continue-2

Test case 2	Low Data Rate (1.1Mbit/s) Solution		(Test Case 2)		Owner: STI
Channel	Current SH1.0 6 channel	Allegro 492 DECT radio 6 channel SH1.0 mode	Test Configs, Chans	Expected Outcome	Status
1	1395.8977	1395.8977	1, 2, 3, 4	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
2	1397.4970	1397.4970	1, 2, 3, 4, 5, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
3	1399.0963	1399.0963	2, 3, 4, 5, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
4	1427.8979	1427.8979	1, 2, 3	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
4a (**)	1430.2410	1430.2410	4, 5, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
5 (*)	1429.4972	1429.4972	3, 4, 5, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
6 (*)	1431.0965	1431.0965	3, 4, 5	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
---	---	---	2, 4a, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
---	---	---	2, 4, 6	MX40 & Allegro Device connects, wave at central	Covered by STI DVR
Current band allocation 1395 to 1400					
Current band allocation 1427 to 1432					
Note: (*) not available in special geographic areas, (**) available in special geographic areas only. and also not available for new sites due to FCC rules					
Rationale: The combination of variables, 7! = 5040, too large for manual test, the combination selected configuration shall not all include ch.1 to test against prior issue. Other, channel combinations can be used and shall be recorded in the STI test case.					

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**7.3 Mixed System level Test Plan-SH2 – Final Test Details**

SH2.0 LAPC (6.x.x.x) Mixed System Functional Performance,

The following verification are performed on Smart Hopping System2.0

SH2.0 Mixed System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.2	Final Acceptance Criteria
System R&D Test and Sanmina Test (R&D/Sanmina) Test case 1.2, 1.3 Test case 4 Test case 5 Test case 6 Test case 7 Test case 8 Test case 9 Test case 10 Test case 11	Test: - R&D Integration Test to confirm basic performance Sanmina Tests: - System Functional Performance - Roam Clients across AP/RA - Roam Clients across Mixed AP (D.03) & SH2.0 AP - Roam Clients (Bearer handover, Connection handover, includes Mixed APs) - System (may include automation) Testing of APC, includes APC web, devices & groups - Layer2, Layer3 networking test - New MAC address detection - DHCP Server for New MAC - Duplicate PMID detection (AP, APC) - New Alerts (Incompatible AP Alert) - Duplicate PMID Alert - Safety Risk SRA - System (may include automation) Testing of APC functions -Using Upgrader tool feature - Scan System and Check APC Configuration - APC: Force Upgrade - APC: Check and Export APC Configuration - AP: Force Upgrade Note: R&D System Testing may leverage applicable elements from prior test plans: 1. Sanmina APC Software Test Plan, PHI-S-0011-APC (Windchill doc# D002051098) ... updated with New MAC test 2. Philips Smart Hopping 1.0 RadioHead AP New MAC Software Test Plan, PHI-S-0012-AP (Windchill doc# D002029403) - Sanmina AP Test Plan	Meets System SRS Results report in Sanmina PHI-S-0012-AP Software Test Report And Sanmina PHI-S-0011-APC Software Test Report

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 Mixed System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.2	Final Acceptance Criteria
Test Case 3 System-Software Test (STI) DECT Paging	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, AP D.03, LAPC 6.x.x.x Find Device (leverage the DECT page function) 1.1 Use Find device feature from PICiX 1.2 Devices, Allegro and MX40 shall respond to Find device. 1.3 Confirm function with Allegro and MX40 Standby/Resume Device (leverage the DECT page function) 2.1 Use standby/resume feature from PICiX 2.2 Devices, Allegro and MX40 shall respond to standby/resume function 2.3 Confirm function met spec for Allegro and MX40 Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A01, System CPE Design Verification Report (LAPC with Mixed SH2.0 systems)
Test Case 4, Test Case 5 APC Premiership & APC Partnership (STI)	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, AP D03, LAPC 6.x.x.x APC Premiership & APC Partnership - Devices, Allegro and MX40 - Initiate Change of APC Premiership - Confirm function with Gen3 AP, and LAPC 6.x.x.x and alert behavior Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A01, System CPE Design Verification Report (LAPC with Mixed systems)
Test Case 6 Layer 2/3 networks (STI)	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, AP D03, LAPC 6.x.x.x Configure Layer 2/3 networks AP D.03, LAPC 6.x.x.x Devices, Allegro (4.x.x.x) and MX40 (C.03.05) - Confirm function with Gen3 AP, and LAPC 6.x.x.x with PICiX	All functions operate per SRS Results in SH2.0_DVR_A01, System CPE Design Verification Report (LAPC with Mixed systems)

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 Mixed System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.2	Final Acceptance Criteria
	Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan	
Test Case 7 APC Web pages (STI)	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, AP D03, LAPC 6.x.x.x a) Generate Duplicate PMID alert with Allegro Client 1) Confirm Individual AP Alert Statistics Web page 2) Confirm Group AP Alert Statistics Web page 3) Confirm function with Gen3 AP, and LAPC 6.x.x.x Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A01, System CPE Design Verification Report (LAPC with Mixed SH2.0 systems)
Test Case 8 System-Software Test (STI) APC DHCP Server, update web page	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) SH2.0 AP, AP D03, LAPC 6.x.x.x APC DHCP Server, update web page 1) Confirm function with Gen3 AP, and LAPC 6.x.x.x 2) Confirm DHCP Server function with Allegro Client and MX40 Client. 3) Include DHCP server failure for SRA risk mitigation ITS-19 (D1, D2) Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A01, System CPE Design Verification Report (LAPC with Mixed SH2.0 systems)
The following Allegro configuration setting is used generally in above test cases: <u>Test Allegro GFSK-Single Slot mode and Encryption Off</u> - At Allegro, go to OpModes → Service, passcode = 273441 - At Allegro service mode, go to Setup Network → Auth/Encryption → Off, - Power cycle Allegro		

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP**7.4 Sample Size (Mixed Systems)**

Sample Plan (Mixed Systems)

7.4.1 AP Test –Sample Size

This applies to RH AP (D.03) and SH2.0 AP, defined as AP below

AP Test	Owner	Device type	Sample size	Sample Size Rationale
AP handover & Roaming Test	R&D	1.4 AP (D.03)	2 devices	Sample size is sufficient for a comparative test (hypothesis analysis is done to confirm the sample mean is equivalent to Legacy AP test mean to accuracy 2dB with 95% confidence)
		SH2.0 AP	2 devices	
		RA	2 devices	This parametric test allows sample mean and variance analysis for the defined interval where 95% confidence is used in this analysis.
		MX40	32 devices	
		Allegro	32 devices	
		SH2.0 Upgrader	1 device	
Compatibility Test	STI	1.4 AP (D.03)	6 devices	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.
		SH2.0 AP	6 devices <i>(1 device type for software)</i>	
System Test	R&D, STI	1.4 AP (D.03)	6 devices	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.
		SH2.0 AP	6 devices <i>(1 device type for software)</i>	
				System includes 24 real devices, as part of Scale Testing

7.4.2 APC Test –Sample Size

This applies to RH LAPC 6.X.X.X and SH2.0 APC, defined as APC below

APC Test	Owner	Device type	Sample size	Sample Size Rationale
Compatibility Test	STI	LAPC	9 devices	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.
			<i>(1 device for software)</i>	
System Test	STI	LAPC	9 devices	This is software test, 1 sample is required per configuration, software function does not vary with

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

APC Test	Owner	Device type	Sample size	Sample Size Rationale
			(1 device for software)	devices. The samples selected provide efficient testing for multiple configurations at the same time. System includes client device emulation for 1024 device support, with 32 real devices. System emulation of 600 APs includes 6 real APs. System with 9 APC

7.4.3 Large Scale System –Sample Size

This applies to LAPC, RH LAPC 6.X.X.X, defined as APC below

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
Devices in System Test ² (System Scale Testing)	STI	MX40, 1.4SH Allegro PWM ITS Bedsides with (990 simulated Clients) PICiX Central Station Focal Point	32 devices 32 devices 2 Bedsides with ITS radio 2 PICiX 1 Focal Point	32 devices sample are required to full load a single AP. Configuration all max AP capacity testing. System includes 32 real devices System includes 2 real ITS Bedside For System Scale Testing Simulated devices are used to reach 1024 devices for system claim. Simulated Devices = 1024 - 34 = 990 System Data loss performance 90 sec/day observed over 24 hours for 32 devices, 95% Confidence/ 91.1% success as an attribute. As a parametric test, with 32 devices Data loss Limit = mean + k1*stdev, k1=2.186 for 95% confidence/ 95% proportion
RAs in System Test	STI	1.4SH Remote Antenna	4 RA	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. AP with 2 Remote Antenna (2 RAs) is the full configuration of software features. AP with No RA is supported.

² All supported devices are included in the system for the integrated system test.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
				Simulated AP = $600 - 12 = 588$ System includes 32 real devices Simulated Devices = $1024 - 32 = 992$
APs in System Test (System Scale Testing)	STI	1.4 AP (D.03) SH2.0 AP	6 AP devices 6 AP devices 4 RA (with 588 simulated APs)	This is software test, 1 sample system is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. System includes 34 real devices (32 PWM & 2 Bedsides) Simulated Devices = $1024 - 34 = 990$ Simulated AP = $600 - 12 = 588$ AP sim
APC in System Test (System Scale Testing)	STI	APC	9 APC devices (1 device for software)	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. System with 9 APC is the max configuration (includes redundancy) System includes client device emulation for 1024 device support, with 32 real devices. System emulation of 600 APs includes 24 real APs.
System Upgrade/Downgrade Reliability	STI	1.4AP (D.03), SH2.0 AP LAPC	6 AP devices 6 AP devices 9 APC devices 1 SH2.0 Upgrader	This is a system software test; this require 1 large scale system. 12 Real APs, 9 APC, 2 RA, 1 Upgrader For reliability testing confirm Upgrade and downgrade has a 99% success with 95% confidence. This is an attribute test. $N = \ln(1-CL)/\ln(P)$ $N = \ln(1-0.95)/\ln(0.99) = 299$ Trials accepting no failures, alternatively The possible execution test trials for upgrade/downgrade with 95% Confidence / 99% Success with accepted failures: 299 trials with 0 failure 473 trials with 1 failure 628 trials with 2 failures 773 trials with 3 failures

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
				Test automation is done within the Upgrader Tool to cycle through a defined number of test trials.
System Scan Reliability	STI	1.4AP (D.03), SH2.0 AP LAPC	6 AP devices 6 AP devices 9 APC devices 1 SH2.0 Upgrader	This is a system software test; this requires 1 large scale system. 12 Real APs, 9 APC, 2 RA, 1 Upgrader For reliability testing confirm system scan has a 99% success with 95% confidence. This is an attribute test. The possible execution test trials for system scan are as follows: 299 trials with 0 failure 473 trials with 1 failure 628 trials with 2 failures Test automation is done within the Upgrader Tool to cycle through a defined number of scan test trials.
For general software reliability test of AP/APC, the target test is to demonstrate statistical performance 99% Confidence/ 99% Prop. This represents a High Reliability solution, a goal for Telemetry Infrastructure. The possible execution attribute test trials for 99% Confidence/ 99% Prop with accepted failures: 459 trials with 0 failure 661 trials with 1 failure 838 trials with 2 failures This type of testing is only practical with automation applied. Attribute Sampling derived from the following $1 - C = \sum_{i=0}^f \binom{n}{i} (1 - R)^i R^{n-i}$ where: C is the test confidence level R is the reliability to be demonstrated f is the number of allowable test failures n is the test sample size				

7.5 SH2.0 System level Test Plan-SH2 – Final Test Details

SH2.0 LAPC (6.x.x.x) & SH2.0 AP System Functional Performance,

The following verification are performed on Smart Hopping System2.0

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.3	Final Acceptance Criteria
System R&D Test and Sanmina Test (R&D/Sanmina) Test case 1.2, 1.3 Test case 4 Test case 5 Test case 6 Test case 7 Test case 8 Test case 9 Test case 10 Test case 11	Test: - R&D Integration Test to confirm basic performance Sanmina Tests: - System Functional Performance - Roam Clients across AP/RA - Roam Clients across SH2.0 AP - Roam Clients (Bearer handover, Connection handover) - System (may include automation) Testing of APC, includes APC web, devices & groups - Layer2, Layer3 networking test - New MAC address detection - DHCP Server for New MAC - Duplicate PMID detection (AP, APC) - New Alerts (Incompatible AP Alert) - Duplicate PMID Alert - Safety Risk SRA - System (may include automation) Testing of APC functions -Using Upgrader tool feature - Scan System and Check APC Configuration - APC: Force Upgrade - APC: Check and Export APC Configuration - AP: Force Upgrade Note: R&D System Testing may leverage applicable elements from prior test plans: 1. Sanmina APC Software Test Plan, PHI-S-0011-APC (Windchill doc# D002051098) ... updated with New MAC test 2. Philips Smart Hopping 1.0 RadioHead AP New MAC Software Test Plan, PHI-S-0012-AP (Windchill doc# D002029403) - Sanmina AP Test Plan	Meets System SRS Results report in Sanmina PHI-S-0012-AP Software Test Report And Sanmina PHI-S-0011-APC Software Test Report
Test Case 3 System-Software Test (STI) DECT Paging	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0AP, LAPC 6.x.x.x Find Device (leverage the DECT page function) 1.1 Use Find device feature from PICiX 1.2 Devices, Allegro and MX40 shall respond to Find	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.3	Final Acceptance Criteria
	device. 1.3 Confirm function with Allegro and MX40 Standby/Resume Device (leverage the DECT page function) 2.1 Use standby/resume feature from PICiX 2.2 Devices, Allegro and MX40 shall respond to standby/resume function 2.3 Confirm function met spec for Allegro and MX40 Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	
Test Case 4, Test Case 5 APC Premiership & APC Partnership (STI)	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, LAPC 6.x.x.x APC Premiership & APC Partnership - Devices, Allegro and MX40 - Initiate Change of APC Premiership - Confirm function with Gen3 AP, and LAPC 6.x.x.x and alert behavior Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)
Test Case 6 Layer 2/3 networks (STI)	Function defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, LAPC 6.x.x.x Configure Layer 2/3 networks SH2.0 AP, LAPC 6.x.x.x Devices, Allegro (4.x.x.x) and MX40 (C.03.05) - Confirm function with Gen3 AP, and LAPC 6.x.x.x with PICiX Test System with MX40 devices (C.03.05) and Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.3	Final Acceptance Criteria
Test Case 9, Test Case 10, Test Case 11 System- Application Software Test (STI)	AP, APC, SH2.0 Upgrader integrated into large system (network Layer2, Layer3), Application functions test as defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, LAPC 6.x.x.x - Covers system function per SRS - Covers supported Network Topologies - Covers APC mastership change - Covers Compatibility per SRS - Covers SRA risks Test System with MX40 devices (C.03.05) Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)
Test Case 12 System- Application Software Test (STI) Compatibility	AP, APC, SH2.0 Upgrader integrated into large system (network Layer2, Layer3), Application functions test as defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, LAPC 6.x.x.x - Covers Compatibility per SRS - SH2.0 AP (<6.x.x.x) with LAPC (6.x.x.x) - SH2.0 AP 6.x.x.x with LAPC (6.x.x.x) - Compatibility 6.x.x.x with PICiX versions (check both patched and un-patched software) - Compatibility with FocalPoint (> A.02) - MX40 and Allegro, Bedsides (SH1.0, SH2.0), Encryption Off - SH2.0 Upgrader 6.x.x.x Test System with MX40 devices (C.03.05) Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)
Test Case 13, Test Case 14 System- Application Software Test (STI) Export, Import	AP, APC, SH2.0 Upgrader integrated into large system Application functions test as defined by 867214-90006 SRS (System Requirements Specification Smart Hopping 2.0) for SH2.0 AP, LAPC 6.x.x.x - Use SH2.0 Upgrader Tool function per SH2.0 Upgrader Service Tool SRS to export system configuration and import system configuration	All functions operate per SRS Results in SH2.0_DVR_A02, System CPE Design Verification Report (LAPC with SH2.0 systems)

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

SH2.0 System Summary verification	What to be Tested, Product Level with Mixed System Configuration Fig.3	Final Acceptance Criteria
	- Covers Export of System Configuration - Covers Import of System Configuration to APCs (Primary, Secondary, Slaves) Test System with MX40 devices (C.03.05) Test System with Allegro devices (4.x.x.x), Encryption Off Covered by SH2.0_DVP, System CPE Design Verification Plan (LAPC)	
The following Allegro configuration setting is used generally in above test cases: 2. <u>Test Allegro GFSK-Single Slot mode and Encryption Off</u> - At Allegro, go to OpModes → Service, passcode = 273441 - At Allegro service mode, go to Setup Network → Auth/Encryption → Off, - Power cycle Allegro		

7.6 Sample Size (SH2.0 Systems)**7.6.1 AP Test –Sample Size**

This applies to SH2.0 AP, defined as AP below

AP Test	Owner	Device type	Sample size	Sample Size Rationale
AP handover & Roaming Test	R&D	SH2.0 AP RA MX40 Allegro SH2.0 Upgrader	4 devices 2 devices 2 devices 32 devices 32 devices 1 device	Sample size is sufficient for a comparative test (hypothesis analysis is done to confirm the sample mean is equivalent to Legacy AP test mean to accuracy 2dB with 95% confidence) This parametric test allows sample mean and variance analysis for the defined interval where 95% confidence is used in this analysis.
Compatibility Test	STI	SH2.0 AP	12 devices (1 device type for software)	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.
System Test	R&D, STI	SH2.0 AP	12 devices (1 device type for software)	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

AP Test	Owner	Device type	Sample size	Sample Size Rationale
				System includes 24 real devices, as part of Scale Testing

7.6.2 APC Test –Sample Size

This applies to LAPC 6.X.X.X and SH2.0 APC, defined as APC below

APC Test	Owner	Device type	Sample size	Sample Size Rationale
Compatibility Test	STI	LAPC	9 devices <i>(1 device for software)</i>	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time.
System Test	STI	LAPC	9 devices <i>(1 device for software)</i>	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. System includes client device emulation for 1024 device support, with 32 real devices. System emulation of 600 APs includes 6 real APs. System with 9 APC

7.6.3 Large Scale System –Sample Size

This applies to LAPC, LAPC 6.X.X.X, defined as APC below

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
Devices in System Test ³ (System Scale Testing)	STI	MX40, 1.4SH	32 devices	32 devices sample are required to full load a single AP. Configuration all max AP capacity testing.
		Allegro PWM	32 devices	System includes 32 real devices
		ITS Bedside	2 Bedside with ITS radio	System includes 2 real ITS Bedside
			with (990 simulated Clients)	For System Scale Testing Simulated devices are used to reach 1024 devices for system claim. Simulated Devices = 1024 - 34 = 990
			2 PICiX	

³ All supported devices are included in the system for the integrated system test.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
		PICiX Central Station Focal Point	1 Focal Point	System Data loss performance 90 sec/day observed over 24 hours for 32 devices, 95% Confidence/ 91.1% success as an attribute. As a parametric test, with 32 devices Data loss Limit = mean + k1*stdev, k1=2.186 for 95% confidence/ 95% proportion
RAs in System Test	STI	1.4SH Remote Antenna	4 RA	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. AP with 2 Remote Antenna (2 RAs) is the full configuration of software features. AP with No RA is supported. Simulated AP = 600 - 12 = 588 System includes 32 real devices Simulated Devices = 1024 - 32 = 992
APs in System Test (System Scale Testing)	STI	SH2.0 AP	12 AP devices 4 RA (with 588 simulated APs)	This is software test, 1 sample system is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. System includes 34 real devices (32 PWM & 2 Bedsides) Simulated Devices = 1024 - 34 = 990 Simulated AP = 600 - 12 = 588 AP sim
APC in System Test (System Scale Testing)	STI	APC	9 APC devices (1 device for software)	This is software test, 1 sample is required per configuration, software function does not vary with devices. The samples selected provide efficient testing for multiple configurations at the same time. System with 9 APC is the max configuration (includes redundancy) System includes client device emulation for 1024 device support, with 32 real devices. System emulation of 600 APs includes 24 real APs.

For **general software reliability** test of AP/APC, the target test is to demonstrate statistical performance 99% Confidence/ 99% Prop. This represents a High Reliability solution, a goal for Telemetry Infrastructure.

The possible execution attribute test trials for 99% Confidence/ 99% Prop with accepted failures:

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

System: AP, APC, Upgrader, Devices Test	Owner	Device type	Sample size	Sample Size Rationale
459 trials with 0 failure 661 trials with 1 failure 838 trials with 2 failures This type of testing is only practical with automation applied. Attribute Sampling derived from the following $1 - C = \sum_{i=0}^f \binom{n}{i} (1 - R)^i R^{n-i}$ where: C is the test confidence level R is the reliability to be demonstrated f is the number of allowable test failures n is the test sample size				

8 Resources

The required verification resources:

- Locations: Cambridge Site, BBN site is optional for Bedsides
- Facilities: STI Lab and R&D Lab
- Human resources (STI Tester, R&D Tester, optionally STI can integrate BBN Bedsides tested)
- Data recording: Record Results into the STICRs
- Azure for DVIP test records. And Windchill for all other test records.

9 Entry Criteria

This is a regression test for released products with new features. The software Transmittal for the fixed STICRs is the key entry requirement.

Entry criteria to consider are:

- The test setup including the used tools for verification must be validated and test equipment must be calibrated.
- Availability of baselined product configurations (physical configuration audit)
- Availability of released Product verification protocols
- Tools and equipment used for verification are validated and calibrated
- No open development problems or a rationale is available
- Trained staffing available

STABILITY OF SOLUTION
Zero Open Enhancement requests or requirements, or pending scope decisions
Zero open defects with severity of Critical, Serious
All features are considered Implementation Complete (For Example: Code checked in/reviewed, unit test complete, integration test complete)

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

Zero open defects identified by SW Test Engineering team as blocking (functionality inaccessible where there is no workaround)
Zero open defects identified by SW Test Engineering team as severely impacting testing (specific feature or broad set of features)
Total number of open defects does not represents an unstable solution environment. (Note the threshold for documentation vs code may be different.)
Decreasing rate of discovery (Consider Features)
Candidate build successfully passes smoke/acceptance test.
R&D has defined migration/upgrade paths and has executed them at least once.
TESTABILITY
Test Concepts and test cases are complete.
Test cases have been validated by dry run or other means.
Unit and Integration Testing is complete and identified defects are recorded and classified.
System Requirements (HW and SW) are defined and all required test systems are available
Test tools are available and accounted for in the Software Tool Inventory and validated as required.
OTHER
Learning products service documentation is available for reference
Updated and approved design inputs Product Requirements/SRS/ SRA are available and under document control.
Shipping worthy installation, configuration mechanisms are documented and available.

10 Exit Criteria

The Exit criteria are:

- Completion of all planned verification activities
- STICRs verified and Closed (Released test records)
- Defects resulted from verification activities shall be in end state or rationale is available

Evidence that acceptance criteria are met (in case defined in the verification strategy)

RESOLUTION OF CHANGE REQUESTS
All defects are closed or dispositioned to another release.
There are zero Critical/Serious defects closed with reasons Not To Be Fixed, Not Reproducible, Duplicate, As specified, Deferred, Transferred.
Cumulative risk of residual defects is acceptable as determined by cross-functional team, as this is STICR testing no cumulative risk is expected and concluded in the Master verification report.
TEST EXECUTION
All STICR Test are executed and passed as indicated by Product Verification Report.
All other regression testing is complete.
OTHER
All solution development activities are complete.

11 Suspension and resumption criteria

The verification suspension and resumption criteria for the verification activities are defined below.

The verification suspension criteria.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

- High number of defect inflow during verification test, such that a significant part of the planned test activities cannot be performed, without running into the consequences of these problems.
- Operational problems, e.g. with infrastructure, test environment, or resources, such that a significant part of the test is blocked.

The verification resumption criteria.

- The defects or operational problems that caused test suspension have been resolved.

The impact of the suspension reason needs to be reviewed on the executed test protocols before resumption. In case the outcome of the executed test protocols is not valid anymore, they need to be repeated

12 Defect Management

The post release defect management process A-Q2920-01310 is used to manage defects. STICRs are created in defect management database and clarified by the cross-functional team. During the development activity for this specific solution if new defects are found the Cross Functional team checks if issue is new to this software build or exists in the released product and is included in the defect classification to manage the issue appropriately.

13 Test Metrics

N/A

14 References

REF	Document Number	revision	Title	Location
Ref-01	867214-90006	≥ K	System Requirements Specification for Smart Hopping 2.0 (Project Trident)	Windchill doc# D000607928
Ref-02	867214-90021	≥ L	Safety Risk Assessment- Smart Hopping 2.0 Infrastructure	Windchill doc# D000834981
Ref-03	867216-90002	≥ J	Smart Hopping 2.0 1.4 AP Software Feature Specifications	Windchill doc# D000607973
Ref-04	867214-90008	≥ H	Smart Hopping 2.0 APC Software Application Feature Specifications	Windchill doc# D000607975
Ref-05	867214-90011	≥ J	Smart Hopping 2.0 Upgrader Software Feature Specifications	Windchill doc# D000870355
Ref-06	A-M4840-90176	L.04.44	Animation Telemetry System Requirements Specification (SRS)	Windchill doc# D001942240
Ref-07	M3185-90050	K.0	Safety Risk Assessment- Smart Hopping Infrastructure	Windchill doc# D001951513
Ref-08	867214-90205	A	Philips New Client MAC Address Support for RadioHead AP/APC and Trident APC- Internal Spec	Windchill doc# D002022790
Ref-09	Dxxxxxxx (SH2.0_MSVP_A01)	None	SH2.0 MAC Address Master System Verification Report (LAPC)	Windchill eDHF
Ref-10	D002072203 (SH2.0_DVP)	A	SH2.0 MAC Address CPE Verification Plan (LAPC)	Windchill doc# D002072203

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

REF	Document Number	revision	Title	Location
Ref-11	Dxxxxxxx (SH2.0_DVR_A01)	None	SH2.0 MAC Address CPE Verification Report (LAPC, mixed SH2.0 system)	Windchill eDHF
Ref-12	Dxxxxxxx (SH2.0_DVR_A02)	None	SH2.0 MAC Address CPE Verification Report (LAPC, SH2.0 system)	Windchill eDHF
Ref-13	PHI-S-0011-APC	external	Philips Smart Hopping 2.0 APC Software Test Plan	Windchill doc# D002043244
Ref-14	PHI-S-0011-APC-R	external	PHI-S-0011-APC-R-6011 SH2.0 APC 6.0.1.1 NewFeature-Regression SW Test Report RevA	Windchill doc# D002063119
Ref-15	PHI-S-0012-AP	external	Philips Smart Hopping 1.0 RadioHead AP New MAC Software Test Plan	Windchill doc# D002029403
Ref-16	PHI-S-0012-AP-R	external	PHI-S-0012-AP-R-D0304 SH1.0 AP D.03.04 NewMAC Baseline SW Test Report RevA	Windchill doc# D002063123
Ref-17	PHI-S-0011-UPG	External	Philips Smart Hopping 2.0 Upgrader Software Test Plan	Windchill doc# D002043269
Ref-18	867216-90065 -PED	TBD	Production-Equivalent Device (PED) Deviation Report	Windchill eDHF
Ref-19	A-Q2920-90751	E	Sample Size Guidance for Design Verification	eDMR

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

15 Document Information

15.1 Document Properties

Property	Value
Template Used	A-Q2920-01678-T001
Document Class	see system managing this doc
Tier	see system managing this doc
Element	see system managing this doc

Printed copies are uncontrolled unless authenticated.

SH2.0 MAC Address Master Verification Plan (LAPC)/ SH2.0_MSVP

16 Traceability Map – System SRS to Verification Plans

Traceability to SRS to Test Cases is covered by STI overall Final Traceability, this will be integrated into the System Master Verification Reports defined below.

The Traceability provided in this document is for the initial planning of the testing activities. All specification from the SRS is included in this traceability map for a complete representation for planning.

This also provide initial Traceability planning fir the Hardware Features specification (AP and APC) to associated testing activities. All specification from the Feature Specs are included in this traceability map for a complete representation for planning.

The System Master Verification Report, A-867214-90023-B01, will include the final traceability details to integrate the test reports when the final reports are vaulted in the DHF. The Final traceability specific to this Phase AP release phase.

The System Master Verification Report, A-867214-90023-D01 includes the final traceability details to integrate the test reports when the final reports are vaulted in the DHF. The Final traceability is specific to this Phase LAPC release phase

The System Master Verification Report, A-867214-90023-E01 (the final document suffix may differ based on number of formal reports created in the project), will include the final traceability details to integrate the test reports when the final reports are vaulted in the DHF.

Traceability of SRS to STI Test Cases for software related specifications will be covered by the System Master Verification Report.

This is synced to STI DVR, 867214-90035-C01, Design Verification Report Smart Hopping 2.0 (DVR) ...
Phase AP.

Traceability of SRS to STI Test Cases for software related specifications will be covered by the System Master Verification Report.

This is synced to STI DVR, 867214-90084-D01, Design Verification Report Smart Hopping 2.0 (DVR) ...
Phase LAPC.

Traceability of SRS to STI Test Cases for software related specifications will be covered by the System Master Verification Report.

This will be synced to STI DVR, 867214-90084-E01 (the final document suffix may differ based on number of formal reports created in the project), Design Verification Report Smart Hopping 2.0 (DVR).